

RESEARCH ARTICLE

Preparation and Properties of pH-Responsive Starch-Based Gel Coated Slow-Release Fertilizer

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ABSTRACT

pH-responsive fertilizer is an innovative slow-release fertilizer that can regulate the release of nutrients according to changes in soil pH, which is designed to improve the applicability of fertilizers according to environmental changes during crop growth. It has research and development potential in the field of smart agriculture in the future. This study developed a starch-based pH-responsive hydrogel (SA) using starch (St) as the matrix and acrylic acid (AA) as the functional monomer, which was subsequently employed to encapsulate urea particles for controlled nutrient delivery. The effects of St/AA ratios and crosslinker concentrations on the hydrogel's water retention, absorption capacity, and swelling behavior were systematically investigated. The fertilizer's release kinetics were evaluated in both aqueous and soil environments, with particular emphasis on pH-dependent responsiveness. Experimental results demonstrated that under typical soil conditions (pH 7 and 9), formulations with St:AA = 3:1 and St:MBA = 500:1 exhibited optimal pH-responsive characteristics. These formulations achieved equilibrium cumulative nitrogen release rates of 80.43% and 80.51%, respectively, with sustained release durations extending to 30–35 days. The findings highlight the potential of starch-based pH-responsive hydrogels as effective carriers for smart fertilizer systems in precision agriculture applications.

1 | Introduction

In the field of agriculture, slow- and controlled-release fertilizers can effectively control the rate, pattern, and duration of nutrient release to distribute and supply nutrients during the growth phase of plants [1, 2], this not only enables crops to improve nutrient utilization and thus achieve higher yields [3], but also reduces the pollution of the soil and the atmospheric environment caused by excessive fertilizers and the loss of economic costs [4].

It has gradually become the preferred application of fertilizers in modern times. However, different soil environments can directly affect the release of slow-release fertilizers [5–7]. Among them, soil acidity and alkalinity (pH) play a significant role in the solubility, migration, and effectiveness of fertilizers, which is one of the most important factors affecting the growth of crops and nutrient uptake [8–10]. Therefore, a more refined study of the role of slow-release fertilizers in different environmental factors, as well as the development of pH-responsive smart slow-release

fertilizers that can adapt to a variety of soil pH conditions and precise release of nutrients, will become the main direction of the future research and development of smart fertilizers.

Hydrogel, a material with high water absorption, high water retention, and slow release of water and nutrients [11, 12], has shown great potential for application in the agricultural field [13, 14]. Among them, hydrogel-coated fertilizers are able to encapsulate the fertilizer nutrients inside the reticulation structure of the hydrogel, and through the hydrogel structure design and permeability study to accurately control the release rate of fertilizers [15–17]. Meanwhile, the hydrogel coating materials preparation cost, water retention, good slow release effect, and degradable [18–20]. In addition, hydrogel materials can realize intelligent responses to the external environment through responsive design [21, 22], which can effectively expand and compensate for the slow-release effect of traditional hydrogel-coated slow-release fertilizers in a single environment, thus further realizing intelligent upgrading of fertilizers by changing the slow-release rate through changes in the soil environment [23, 24].

Currently, researchers are deeply exploring the responsiveness of responsive gel materials between different pH environments: Han et al. [25] fabricated capsule switches consisting of two pH-responsive hydrogels in a bilayered structure, which can be bent in alkaline or acidic environments to achieve the “open” motion of the capsule switch, and has been tested for substance release. Similarly, Gao et al. [26] prepared carboxymethyl chitosan pH-responsive hydrogels, and the slow-release performance experiments showed that the cumulative release rates of the target release components in the gel were 29.9%, 49.3%, and 65% at pH = 7.4, 6.5, and 5.0, respectively, which significantly prolonged the release time. In addition, Liang et al. [27] prepared injectable mucus and pH-responsive hydrogels by grafting chitosan with dihydrocaffeic acid and pullulan polysaccharide oxides via the Schiff base reaction and showed significant pH-responsive properties at different pHs (pH = 7.4, pH = 6.8 and 5.5). The above studies on the pH responsiveness of gels are biased towards the application of slow release of medical drugs. As an opportunity, we extended the application of pH-responsive gels to slow-release fertilizers to realize that when the environmental stimulus signals change, the properties of the responsive gels, such as the shape, the permeation rate, the recognition properties, and the release properties, change accordingly, and thus realize the regulation of the release rate [28, 29]. The low-cost gel is selected as the matrix to introduce pH-responsive functional groups, simplify the preparation technology, reduce the preparation cost, and use the pH-responsive gel material as the coating layer to prepare slow-release fertilizers, which intelligently regulate the release of internal fertilizers according to the change of soil pH, so that this intelligent response mechanism can more accurately satisfy the nutrient demand of crops at different growth stages and under different soil conditions.

In this work, common low-cost corn starch was used as a base material and acrylic acid (AA) as a monomer to prepare pH-responsive starch-based hydrogels, which were encapsulated in the outer layer of fertilizer particles to prepare hydrogel-encapsulated slow-release fertilizers. The coated gel-responsive material exhibits different changes in swelling and permeation under different pH environmental conditions through a

protonation or ionization process of a group ($-\text{COOH}$) within the polyacrylic acid that is sensitive to pH changes in the gel. Alterations in the permeability of the gel-coated substance result in the regulated discharge of nutrients from the inner fertilizer, investigating its viability as a coating agent for pH-sensitive delayed-release fertilizers.

2 | Materials and Methods

2.1 | Materials

Corn starch (St) was purchased from Shanghai Macklin Biochemical Technology Co. Ltd. in China; acrylic acid (AA, 99%) was purchased from Tianjin Kemiou Chemical Reagent Co. Ltd. in China, and N,N'-methylenebisacrylamide (MBA, 99%) was purchased from Sinopharm Chemical Reagent Co. Ltd. in China. Potassium persulfate (KPS, 99%) was purchased from Shanghai Aladdin Biochemical Technology Co. Ltd. in China, and sodium hydroxide ($\text{Ca}(\text{OH})_2$, 99%) was purchased from Tianjin Baishi Chemical Co. Ltd. in China; hydrochloric acid (HCl, AR) was purchased from Tianjin Hengxing Chemical Reagent Co. Ltd. in China; urea ($\text{CH}_4\text{N}_2\text{O}$, 99%) was purchased from Tianjin Guangfu Technology Development Co. Ltd. in China. All the water used in this study was distilled water; for special mention, the reagents were analytical grade reagents and could be used without further purification.

2.2 | Preparation of Amorphous Starch

Twenty-five grams of corn starch was added to a three-necked round-bottomed flask containing 100 mL of 50% ethanol by volume and placed in a water bath at 85°C for 2 h with mechanical stirring at 150 rpm. The mixed reaction solution after stirring treatment was diafiltrated and washed repeatedly with 50% volume fraction ethanol to remove impurities and unreacted material. The washed solids were placed in a blast drying oven and dried at 50°C for 24 h. The dried solids were pulverized and sieved through a sieve to obtain amorphous starch powder.

2.3 | Preparation of pH-Responsive Starch-Based Gels (SA)

Weigh 5 g of amorphous starch in a three-necked round-bottomed flask, add 95 mL of distilled water and stir well. Acrylic acid (AA) and N,N'-methylenebisacrylamide (MBA) were weighed at specific molar ratios and added to starch (St) solutions to prepare the gel samples listed in Table 1. These gels were designated as pH@SA-* and pH@SM-*, corresponding to their respective St/AA and St/MBA molar ratios. The molar ratios were calculated based on the molecular weight of the repeating glucose unit ($\text{C}_6\text{H}_{10}\text{O}_5$) in St. The above mixed solution was pre-crosslinked by stirring at 150 rpm in a 60°C water bath for 30 min. Potassium persulfate (KPS) with a molar ratio of St to KPS of 400:1 was weighed and added to 5 mL of water to dissolve, and the reaction mechanism, illustrated in Figure 1, involves the formation of active sites through free radicals generated by KPS pyrolysis. These radicals initiate chain reactions with acrylic acid (AA) via interaction with hydroxyl groups, followed by cross-linking polymerization of the

mixed solution in the presence of N,N'-methylenebisacrylamide (MBA), ultimately resulting in the synthesis of SA gel. After the response, the resulting gel was washed three times in distilled water, put into a vacuum drying oven at 50°C to dry for 48h, and a portion of the gel was ground for other tests.

2.4 | Preparation of pH-Responsive Starch-Based Gel-Coated Slow-Release Fertilizer (SA-SRF)

Spraying 10mL of anhydrous ethanol onto the urea surface ensured uniform wetting. Subsequently, the urea surface was evenly coated with pre-ground SA powder (<100 mesh) using

TABLE 1 | Formulation of pH-responsive starch-based gels (SA).

Sample	Content of starch (St)	Molar ratio of St to AA	St with the MBA mole ratio
pH@SA-1	5 g	2:1	1000:1
pH@SA-2	5 g	3:1	
pH@SA-3	5 g	7:2	
pH@SA-4	5 g	4:1	
pH@SA-5	5 g	5:1	
pH@SM-1	5 g	4:1	300:1
pH@SM-2	5 g		500:1
pH@SM-3	5 g		1500:1
pH@SM-4	5 g		2000:1

the disk method, applying three layers, and then dried at 50°C for 48 h to produce SA-SRF.

2.5 | Structural Characterization of Hydrogels

2.5.1 | Fourier Transform Infrared Spectroscopy Test (FTIR)

Fourier transform infrared spectroscopy (FTIR) is used to determine the absorption spectra of the samples in a particular range, and the samples were milled and pressed into thin slices using the KBr pressing method according to the sample to KBr mass ratio of 1:100, and then passed through the Fourier Transform Infrared Spectrometer (FTIR, IFS 66v/s, Bruker, Germany) in a wavelength range from 400 to 4000 cm⁻¹.

2.5.2 | X-Ray Diffraction Test (XRD)

The prepared hydrogel samples were tested by x-ray diffraction (XRD, D8 Advance, BRUKER, Germany). The test conditions were selected from an angular range of 10°–90°, a scanning speed of 10°/min, a tube voltage of 40 kV, and the use of a copper target for radiation.

2.5.3 | Scanning Electron Microscope Test (SEM)

The pH responsive gel morphology was observed using a scanning electron microscope (SEM, INSPECT F50). Before observation, the SA gel samples of each component were rapidly quenched in liquid nitrogen, followed by freeze-drying to

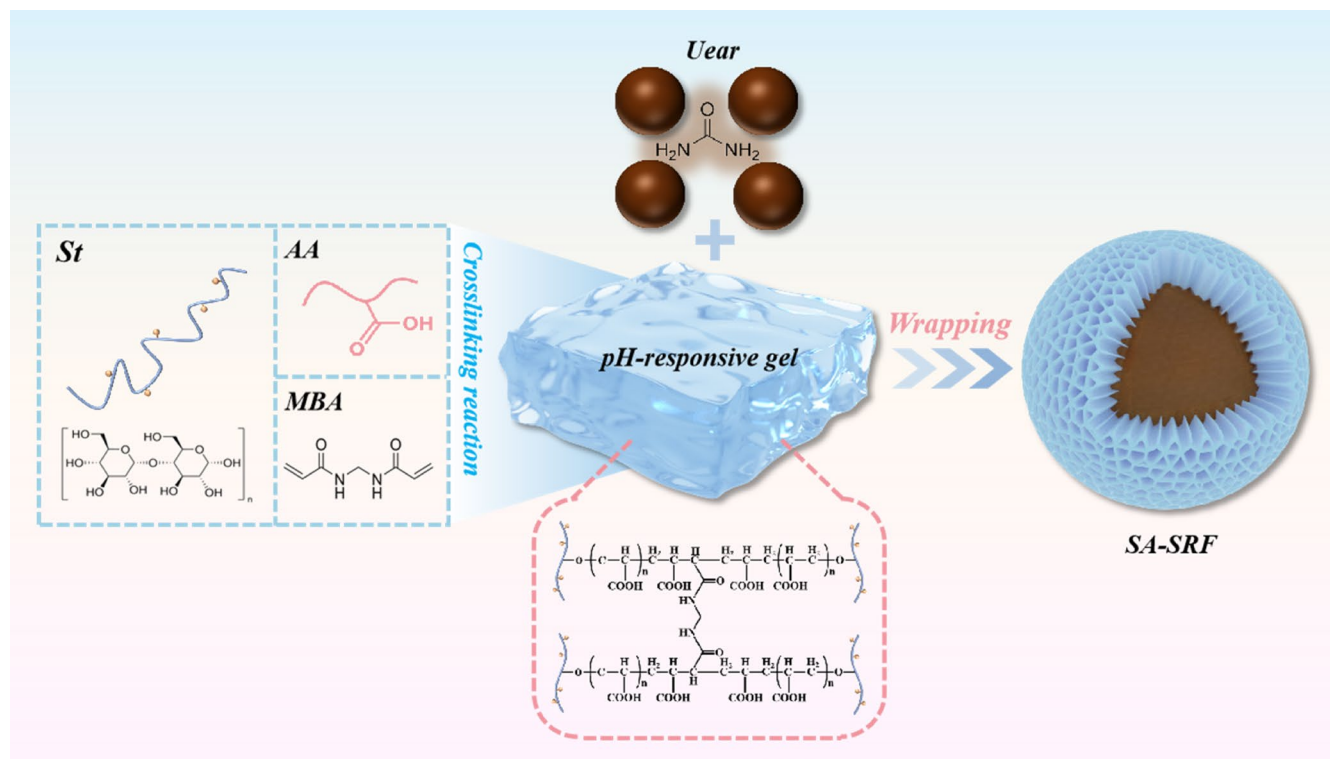


FIGURE 1 | Synthesis mechanism diagram of SA gel as well as SA-SRF. [Color figure can be viewed at wileyonlinelibrary.com]

remove water while preserving the porous structure. The samples were fixed on a copper column using conductive adhesive, and then they were gold sprayed. After the gold spraying was completed, the surface morphology of the samples was characterized by SEM. The test conditions were a voltage of 5 kV and a current of 10 μ A.

2.6 | Determination of Gel Yield and Gel Fraction

The gel yield, calculated using Equation (1), represents the percentage of dry matter content in the freshly prepared gel.

$$\text{Gel yield} = \frac{W_d}{W_n} \times 100 \% \quad (1)$$

where W_d represents the dry weight of the gel after complete dehydration, and W_n corresponds to the initial fresh weight immediately following gel preparation.

The gel fraction of the gel was calculated by the Equation (2) by first measuring the dry weight of the gel, followed by placing the dry gel in a non-woven ziplock bag and adding distilled water, waiting until the gel was completely dissolved, and finally weighing the mass of the dissolved gel after freeze-drying.

$$\text{Gel fraction} = \frac{W_t}{W_i} \times 100 \% \quad (2)$$

where W_t represents the dry mass of the gel prior to dissolution, and W_i corresponds to the mass of the insoluble fraction following complete dissolution and subsequent freeze-drying.

2.7 | Swelling Properties of Gels in Different pH Aqueous Environments

The performance of SA gels in response to pH was analyzed by determining the swelling equilibrium rate of the gels at different pH. Weigh 1 g of the dried hydrogel and denote it as M_d (g); it was submerged in a beaker containing a water environment of pH = 3, 5, 7, 9, 11 (± 0.2) and put into a constant temperature incubator at 25°C. The sample was taken out every 5 h, and the excess water droplets on the surface were absorbed and weighed, denoted M_s (g), and so on until the samples reached the equilibrium of dissolution. The rate of water absorption and swelling of the hydrogel in an aqueous environment at each pH for different time periods was calculated by using Equation (3), denoted as W_s (g/g).

$$W_s (\text{g/g}) = \frac{M_s - M_d}{M_d} \times 100 \% \quad (3)$$

2.8 | Water Retention and Holding Capacity of Fertilizers at Different pH Conditions

The experiments were conducted at a relative humidity of 20%–30% and a temperature of 20°C $\pm$ 5°C. The soil was drenched several times with deionized water and left to stand for 30 min. Repeat the process until it measures its pH = 7 (± 0.2). A small amount of diluted HCl and Ca(OH)₂ solutions was added to each

of the target pH soil samples of the predefined configurations using small, uniform, and gradual sprays, respectively. Measure the pH of the soil leachate and let it rest for 30 min; measurements were taken again, and the above steps were repeated until the soil samples were adjusted to the respective desired pH (± 0.2). The conditioned individual soil samples were left until dry to obtain individual soil samples of the desired pH, which were measured throughout the period using a high precision pH meter. One gram of dried (SA) gel and 200 g of soil of different pH were weighed separately; they were mixed uniformly and placed separately into PVC tubes. The bottoms of the tubes were sealed with non-woven mesh, and the whole was weighed and recorded as (W_0). Add distilled water from the top of the tube, gradually wetting the mixture until water seeps out of the bottom of the tube. When no more water seeped out, the whole was weighed and recorded as W_1 . The soil's water retention capacity ($W_h\%$) was determined using Equation (4).

$$W_h \% = \frac{W_1 - W_0}{200} \times 100 \% \quad (4)$$

Using soil samples prepared in the same way described above, each soil column was weighed (W_d) every 5 days at a temperature of 20°C $\pm$ 5°C after no more water had seeped out of the bottom of the tube. The water retention of the soil, $W_D\%$, was calculated from Equation (5).

$$W_D \% = \frac{W_d - W_0}{W_1 - W_0} \times 100 \% \quad (5)$$

2.9 | Determination of Slow-Release Properties of Fertilizers

Using p-dimethylaminobenzaldehyde (PDAB) as the chromogenic agent, it reacts to a configured standard solution of urea under acidic conditions to form yellow p-dimethylaminoformaldehyde urea, and the absorbance of different urea concentration contents was measured using an ultraviolet spectrophotometer (UV-752N) at 426 nm, plotting the standard absorption curve and fitting the regression equation. The urea content was obtained by absorbance, and further, the release rate profile of fertilizer nutrients in water was obtained.

2.9.1 | Determination of Slow-Release Properties of Fertilizers in Water and in Soil

The nutrient release properties of the samples were determined using the dissolution rate in water method: 1 g of urea and slow-release fertilizer samples were immersed in a beaker containing 100 mL of distilled water (pH = 7) and other aqueous solutions with pH values of 3, 5, 9, and 11; the beakers were capped with plastic wrap for sealing, and the solution in the beakers was sampled at 12 h intervals, taking care to add distilled water after each sampling to maintain a constant solvent volume. Quantitative analysis using an ultraviolet spectrophotometer (UV-752N) was used to determine and calculate the amount of N^{3+} in the samples and to plot release rate curves.

Soils of different pH were prepared according to the methods in Section 2.7 of this paper so that the fertilizers could be tested for

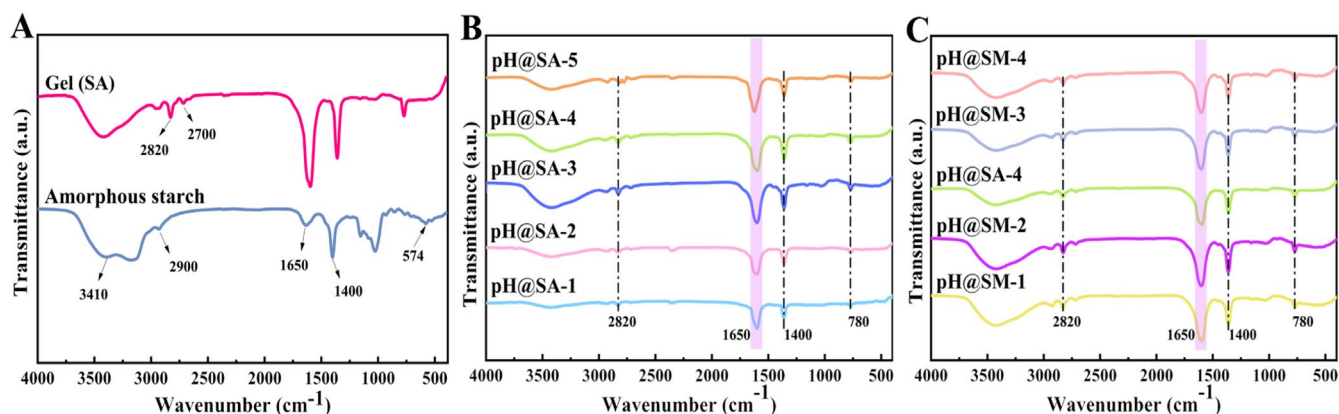


FIGURE 2 | FTIR spectra of starch-based gel SA and amorphous starch (A), and starch-based gel samples with different ratios (B, C). [Color figure can be viewed at wileyonlinelibrary.com]

slow-release properties. It is important to note that soil pH may vary temporally due to inherent buffering capacity during the experimental period. To maintain pH stability, we implemented a rigorous monitoring protocol involving daily measurements using a high-precision pH meter. When pH deviations exceeded ± 0.2 from the target value, we applied minimal volumes of acid or alkali solutions through controlled spraying, followed by continuous monitoring until stable pH conditions (± 0.2) were re-established. For the release experiments, during testing, 1 g of urea and slow-release fertilizer were mixed with soil samples of different pH levels. The mixed soil samples were placed in PVC tubes, and the bottom of the tubes was sealed with non-woven fabric mesh, and according to the soil pH change, timely use of acid or alkaline adjustment to maintain the soil pH state, every 5 days to take its leachate, according to the measurement of leachate absorbance calculation N^{3+} content, so as to calculate the calculation of the release rate curve.

3 | Results and Discussion

3.1 | FTIR Characterization Analysis

The FTIR test results of each component are shown in Figure 2, and the telescopic vibration absorption peaks of the hydroxyl group ($-OH$) and carbonyl group ($C=O$) on the starch molecular chain are at $3200\text{--}3400$ and 1650 cm^{-1} in amorphous starch in Figure 2A, respectively. In the SA gel, 2820 cm^{-1} is the stretching vibration absorption peak of the hydroxyl group ($-OH$) in AA, and the increase in the intensity of the peaks at 1650 and 1400 cm^{-1} is due to the synergistic effect of the carbonyl group ($C=O$) in AA and starch and the antisymmetric and stretching vibration of the carboxyl group in AA. Furthermore, by comparing the absorption peaks of the SA gel and starch, a significant weakening of the absorption peak associated with the cyclic structure of glucose units near 574 cm^{-1} was observed in the SA gel. This demonstrates that the ($C-C$) bonds connected to the hydroxyl ($-OH$) groups in the cyclic glucose units were broken during polymerization. The resulting free radicals initiated the graft polymerization reaction of AA, confirming the successful preparation of the SA gel [30]. As shown in Figure 2B, when the St content gradually increased compared to AA, the intensity of the peaks at 1650 cm^{-1} , and so forth, of the SA gel first increased

and then decreased, and the maximum peak was achieved here at sample pH@SA-3, while the intensity of the peaks at 1650 cm^{-1} showed a gradual increase with the gradual decrease of the MBA content in Figure 2C. This was due to the crosslinking degree of the gel reaching equilibrium.

3.2 | X-Ray Diffraction (XRD) Spectral Analysis

In Figure 3A, the diffraction peaks of ordinary corn starch appear at 2θ of 18.2° , 22.6° , 34.2° , and 81.3° with higher wave intensities of starch, and there are many sharp peaks between 10° and 25° , which are originated from the impurity components in starch. In amorphous starch, the plots show that the baseline is relatively smooth. The diffraction peaks at 18.2° , 22.6° , and 34.2° are relatively broader and weaker in intensity, indicating that the amorphous starch has a significantly lower degree of crystallinity compared to starch, and that this kind of gel prepared from amorphous starch will have a higher degree of crosslinking. In contrast, the SA gel only had a broader and blunter single peak between 2θ of 10° and 25° (Figure 3B,C). With the gradual increase of the molar ratio of St to AA, the intensity of the peaks of the SA gel between 2θ of 10° and 25° decreased and then increased, and the crystallinity of the gel sample pH@SA-2 was the lowest. A similar pattern occurs when St increases relative to MBA content in Figure 3C, in which the baseline of the component plots is relatively smooth.

3.3 | Gel Microstructure Analysis (SEM)

Microstructural analysis of the mesh cell structures formed by freeze-dried SA gels with different St:AA ratios was conducted based on the scanning electron microscope images shown in Figure 4 [31, 32]. In the pH@SA-1 gel, the mesh cells are relatively large with only a few micropores present, and extensive gel skeleton walls are observed, indicating a broad gel framework. In contrast, the pH@SA-2 gel exhibits densely packed and uniformly distributed mesh cells with moderate sizes ranging from 20 to $40\text{ }\mu\text{m}$. The extensive gel skeleton walls gradually thin from thick to thin, leading to larger mesh cells and a significant increase in their number. In the pH@SA-3 sample, mesh cell diameters range from 30 to $50\text{ }\mu\text{m}$, exhibiting larger

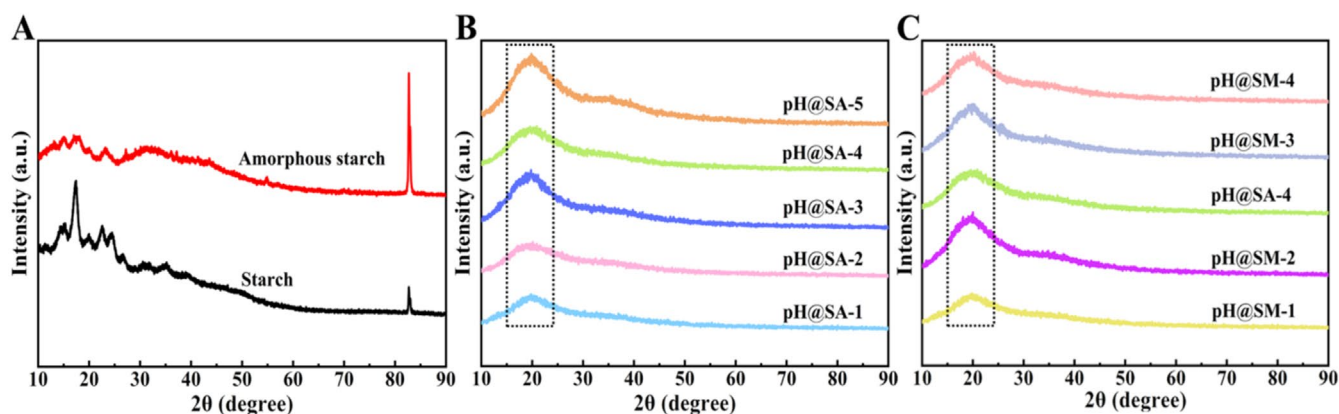


FIGURE 3 | XRD patterns of SA gels of starch with amorphous starch (A), different corn starch and AA molar ratios (B), and different starch to MBA molar ratios (C). [Color figure can be viewed at wileyonlinelibrary.com]

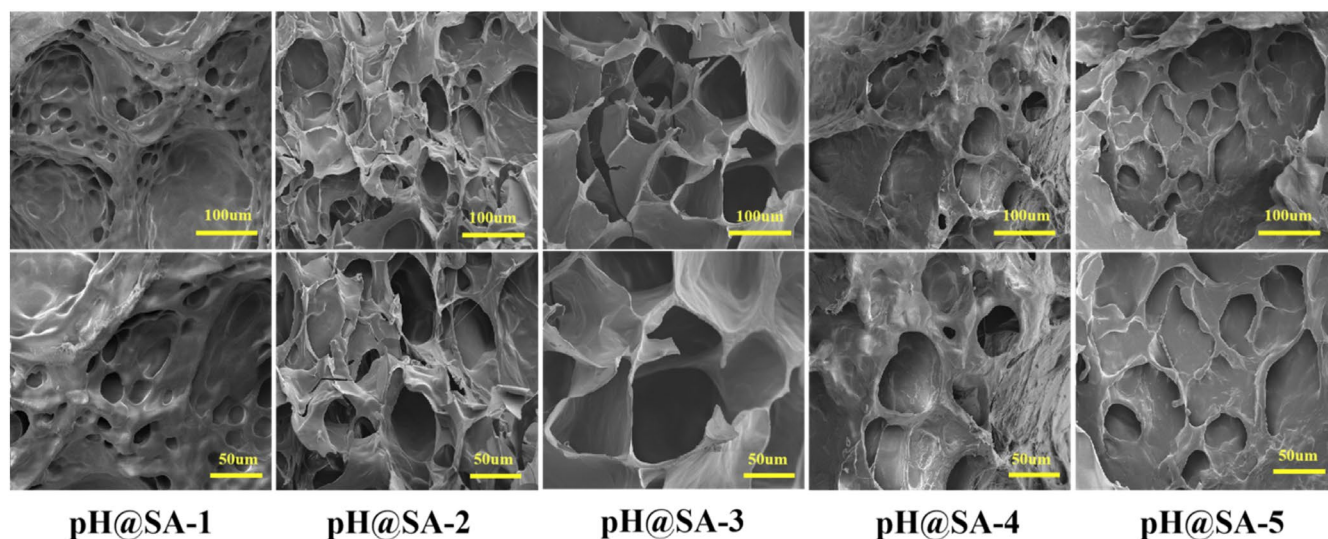


FIGURE 4 | SEM images of freeze-dried gel samples (with different St:AA ratios) at various magnifications. [Color figure can be viewed at wileyonlinelibrary.com]

dimensions compared to pH@SA-2, accompanied by increased gel skeleton wall thickness. Sample pH@SA-4 shows a reduced number of mesh cells and the formation of “vesicular” structures, with a noticeable increase in gel wall thickness. The pH@SA-5 gel exhibited relatively thicker framework walls, with the largest mesh cell dimensions occurring on the framework wall surface. This indicates that as St content gradually increases, the mesh cell diameter of SA gels progressively enlarges, their number first increases then decreases, and the gel walls gradually thicken. Comparative analysis reveals that the pH@SA-2 gel sample possesses the most uniform structure and the highest number of mesh cells.

Figure 5 shows the microstructure of SA gels with different St:MBA ratios after freeze-drying. Observation reveals that pH@SM-1 exhibits a dense three-dimensional network structure with a fine gel skeleton and tightly arranged mesh cells measuring approximately 10 μm in diameter. In contrast, the pH@SM-2 gel exhibits coarser framework elements, with mesh cell diameters increasing to 10–25 μm. Compared to pH@SM-1, this configuration maintains dense mesh cells while providing a more robust gel framework. The pH@SA-4 gel exhibited even

larger mesh cell diameters ranging from 70 to 100 μm, with a gel skeleton width increasing to 20 μm. The pH@SM-3 gel, containing a lower crosslinker content, showed reduced mesh cells and noticeably thickened gel walls. The pH@SM-4 gel, with the lowest crosslinker content, exhibited the largest mesh cell diameter and thickest gel walls. This indicates that as the relative content of MBA decreases compared to St, the mesh cell diameter and gel skeleton width of SA gels gradually increase, the number of mesh cells gradually decreases, and the gel density gradually decreases. The pH@SM-2 gel with a ratio of St:MBA = 500:1 exhibits the most suitable density and regular pore structure.

3.4 | Factors Affecting the Swelling Properties of SA Gels

Figure 6A presents the gel yields of SA gels under different formulations, revealing consistently high and comparable values across all samples. Notably, pH@SA-1, pH@SA-2, pH@SM-1, and pH@SM-2 exhibited the highest yields, approaching or exceeding 4.8%. As shown in Figure 6B, the gel fractions—calculated as the ratio of dry weight before swelling to dry weight

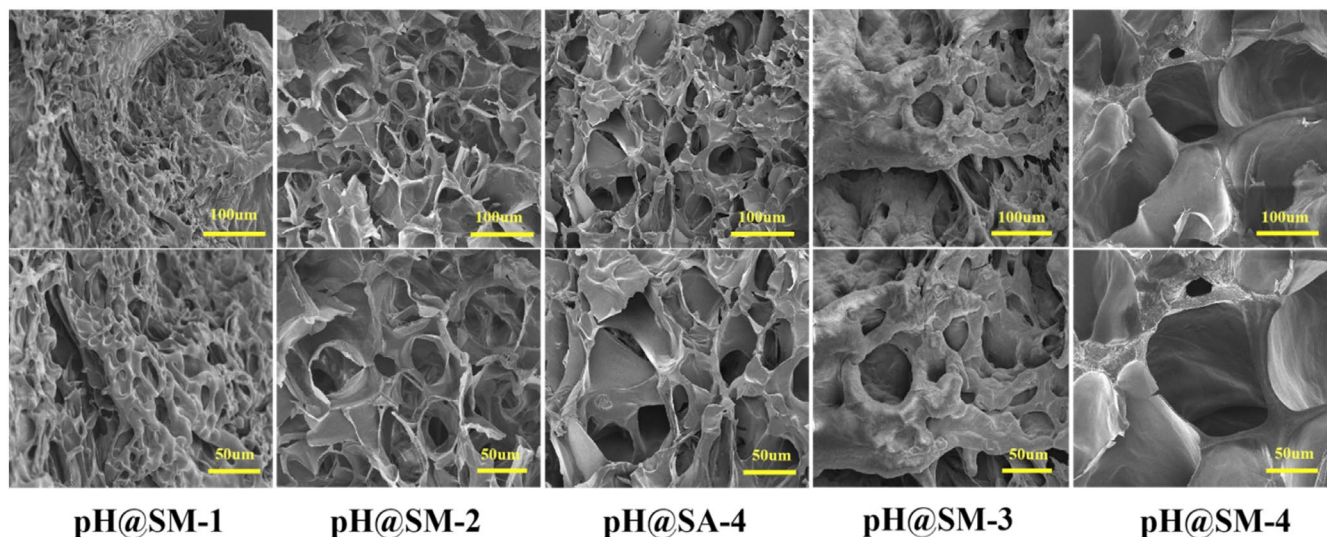


FIGURE 5 | SEM images of freeze-dried gel samples (different St:MBA ratios) at various magnifications. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

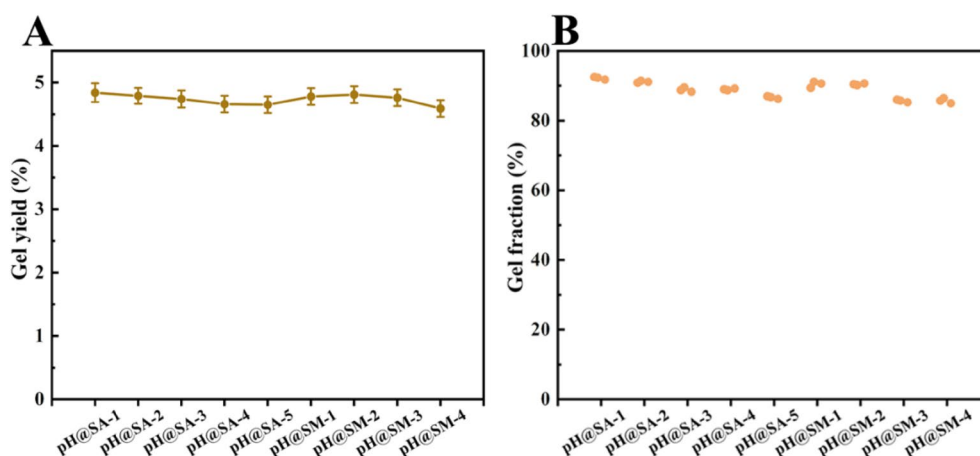


FIGURE 6 | Gel yield and gel fraction of SA gels. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

after swelling equilibrium—remained stable at approximately 90% for all formulations, confirming robust network stability. Consistent with the gel yield trends, pH@SA-1, pH@SA-2, pH@SM-1, and pH@SM-2 displayed the highest gel fractions, attributable to their elevated MBA content (Table 1). This higher crosslinker concentration promoted a denser crosslinking network, thereby enhancing gel stability and swelling resistance, which ultimately contributed to the increased gel fraction.

3.5 | Factors Affecting the Swelling Properties of SA Gels

The swelling curves and equilibrium swelling rates of the gels prepared with different St to AA ratios are demonstrated in Figure 7A,B. The pH@SA-1 gel shown in the figure reached the swelling equilibrium at 110h with the highest equilibrium swelling rate of 7.7g/g, whereas pH@SA-2 exhibited the most considerable equilibrium swelling rate (7.85 g/g) and the fastest time to swelling equilibrium (30h) among all the gels. This is due to the increase in the AA content with hydrophilic groups

in the gel, which improves the hydrophilicity of the gel, and the larger pore size of the gel, so it leads to a faster rate of adsorption of water molecules. This results in speedier adsorption of water molecules, which leads to more excellent water absorption and swelling performance and more rapid absorption and swelling equilibrium time. However, pH@SA-3, pH@SA-4, and pH@SA-5 were not as good as pH@SA-2 gels, with equilibrium swelling rates of 6.3, 5.7, and 5.9 g/g, respectively, due to the increase in the proportion of St and the limitation of the degree of crosslinking, which led to the instability of the gel structure, and the fragmentation and cleavage of gels in the process of swelling, which led to the lower equilibrium swelling rate of gels than that of the previous of the gel components.

Figure 7C,D presents the swelling curves and equilibrium swelling rates of gels synthesized with varying ratios of St and MBA. Through observation, it was determined that pH@SM-4 achieves swelling equilibrium most rapidly, with an equilibrium time of 45 h. However, its equilibrium swelling rate is relatively low, at only 4.7 g/g, and the equilibrium swelling rates of pH@SA-4 and pH@SM-3 are 5.7 and 5.3 g/g,

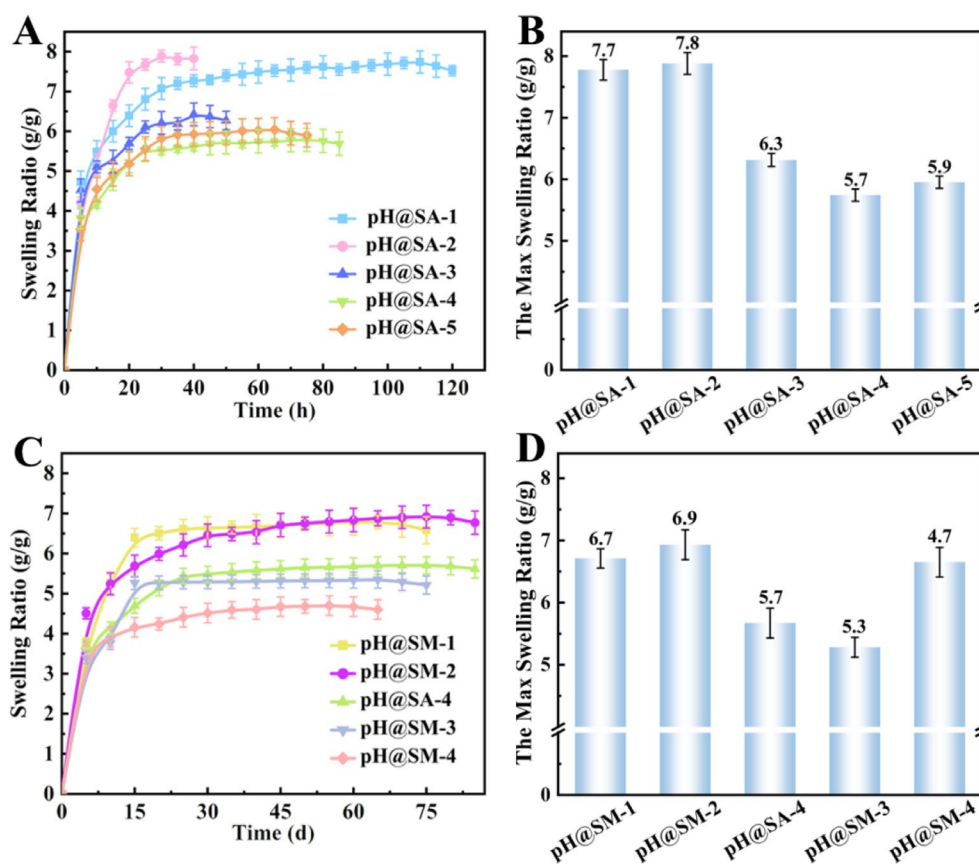


FIGURE 7 | Absorption swelling rate (A, C), equilibrium swelling rate (B, D) for each different SA gel. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.com)]

respectively, which are relatively low. In contrast, the SA gel pH@SM-2, with a St to MBA molar ratio of 500:1, exhibits the highest equilibrium swelling rate (6.9 g/g) and the longest time to reach swelling equilibrium (75 h). This result can be attributed to the fact that when the MBA content is too low, the overall degree of crosslinking within the gel network is insufficient. During the swelling process of the hydrogel, the matrix material with weak crosslinking dissolves into water, leading to a larger pore diameter. Consequently, the water absorption balance within the gel is reached more rapidly, ultimately reducing the overall water absorption capacity of the gel. However, the pH@SM-2 gel with a St to MBA molar ratio of 500:1 possesses a robust structure and moderate pore size. The presence of the AA component enhances its hydrophilicity, thereby resulting in a higher equilibrium swelling rate compared to other gels.

3.6 | Response of SA Gels to pH

The pH responsiveness of St and AA relative content to SA gels is shown in Figure 8A. Among them, the equilibrium swelling of pH@SA-1 and pH@SA-5 increased and then decreased with the gradual increase of pH, but showed better absorptive swelling performance in neutral and weakly alkaline aqueous environments. The pH-responsive curves of pH@SA-2 and pH@SA-3 gels showed an opposite trend, with pH@SA-2 exhibiting superior water-absorbing and swelling properties in all pH solutions. Overall, the corresponding performance trends of the gel pairs

varied across pH environments, but all demonstrated excellent water absorption and swelling capabilities, especially in neutral and weakly alkaline aqueous environments. This unique pH responsiveness in SA gels comes from the action of the internal polyacrylic acid (PAA) chain segments binding or ionizing with protons in the environment: At low pH, $-\text{COOH}$ protonates and hydrophobicity plays a dominant role, leading to the curling of the polymer chains containing carboxyl groups and a decrease in the interchain distance, resulting in the contraction of the pores of the hydrogel network and a decrease in the pore size, thus restricting the passage of permeable media. In contrast, at high pH, $-\text{COOH}$ dissociates into negatively charged $-\text{COO}^-$, and electrostatic repulsion causes the polymer chains to stretch and the chain spacing to increase, leading to network pore expansion and easier passage of permeable media, thus enabling the gel to exhibit responsive properties.

In Figure 8B,C, pH@SA-1 and pH@SA-2 possessed relatively high equilibrium dissolution rates in aqueous environments within a specific pH range, with pH@SA-2 possessing the highest equilibrium dissolution rate (7.9 g/g) of all samples and pHs in aqueous environments at pH = 9. Even at pH = 11, the aqueous environment showed the lowest equilibrium swelling rate still as high as (7.5 g/g), which was significantly higher than pH@SA-1 and the other gels in Figure 8D–F, and all of them had relatively shorter equilibrium swelling times. This is due to having a higher percentage of PAA chain segments in the gel, which allowed the gels to make a better swelling response to the aqueous environments at all pHs. pH@SA-2

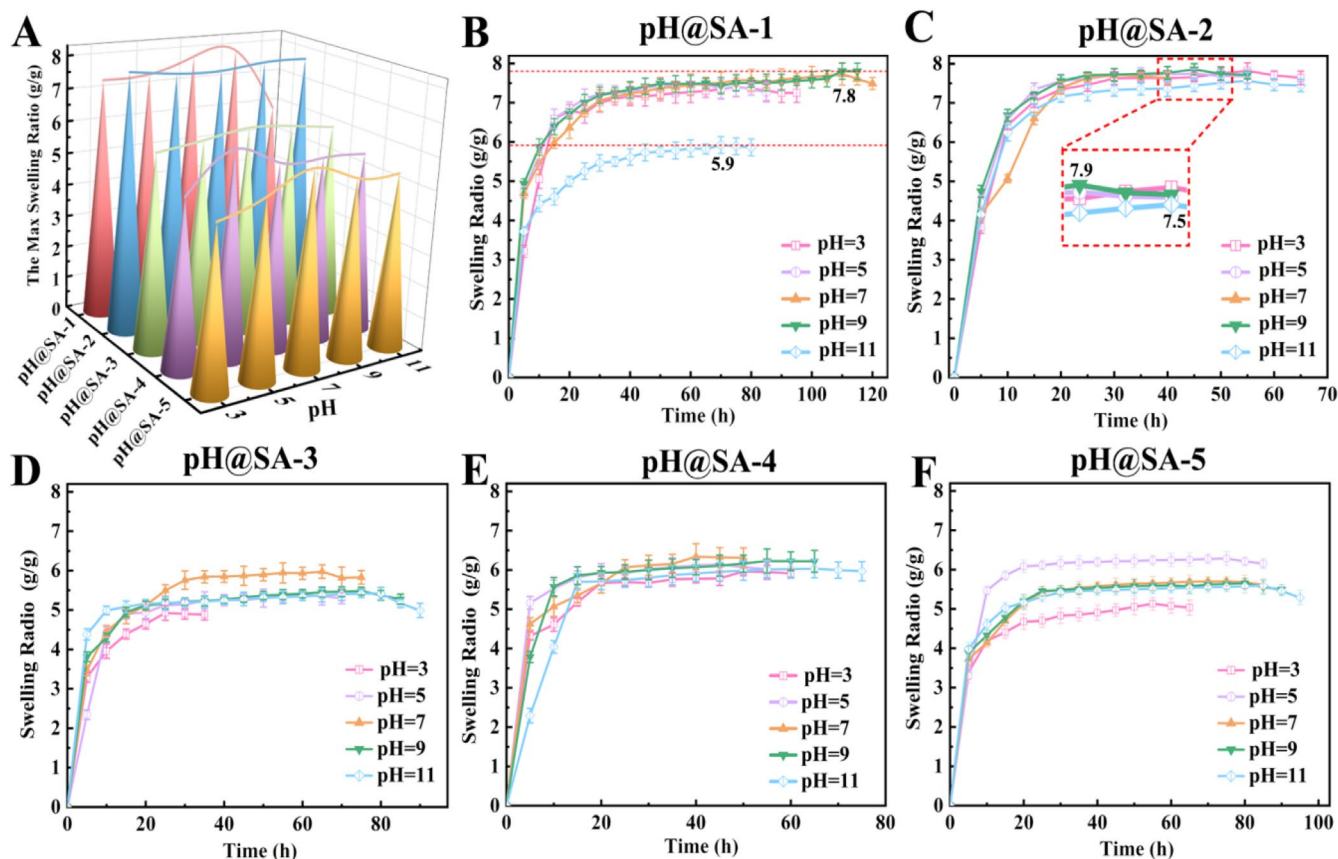


FIGURE 8 | Changes in equilibrium swelling rate response of SA gels with different AA contents at different pH (A) and swelling rate change curves (B–F). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

exhibits low fluctuation and high swelling in various pH environments, which is conducive to ensuring that the gel fertilizer can rapidly draw in and retain water in arid soils with different pH environments.

In Figure 9A, each gel with different crosslinker contents exhibits a more uniform level of equilibrium dissolution rate, but changes differently in response to changes in pH. The more obvious ones, pH@SM-1 and pH@SM-4 gels, had opposite trends of dissolution changes in acidic and strongly alkaline aqueous environments; however, they were the same in neutral and weakly alkaline conditions. The rest of the gels possessed a similar trend of changes, with a gradual increase in strongly acidic environments, a gradual decrease in weakly acidic environments, and then a gradual leveling off in neutral and alkaline environments. By examining the numerical values and observations, it is found that the pH@SM-1 and pH@SM-2 gels exhibited the same highest equilibrium swelling rate of approximately 7.2 g/g in different aqueous environments; however, the changes in the swelling equilibrium values of pH@SM-2 were more stable throughout the different pH changes, as shown in Figure 9B,C. The rate of reaching equilibrium swelling is faster compared to pH@SM-1, compared to the rest of the gels, which have relatively low equilibrium swelling rates at all pH aqueous environments (Figure 9D–F). This is since the increase in the percentage of crosslinker content in the gel leads to the rise in the number and density of gel voids, and the increase in the crosslinking density leads to a higher level of equilibrium swelling rate of

the gel in aqueous solutions with different pH changes and a smooth value.

3.7 | Morphology and Storage Stability of SA-SRF

Figure 10A,B presents comparative SEM micrographs of the SA-SRF gel coating before and after extended storage. Microscopic analysis demonstrates that the gel coating maintains excellent adhesion to the fertilizer core without exhibiting delamination or structural degradation, preserving its original layered architecture throughout the storage period. Furthermore, FTIR spectroscopic analysis (Figure 10C) reveals no detectable changes in the characteristic vibrational modes between freshly prepared and aged samples, with the carbonyl stretching band ($\text{C}=\text{O}$) and carboxyl group vibrations (1650 cm^{-1} , AA) maintaining identical peak positions and intensities. These collective findings provide conclusive evidence that the SA-SRF controlled-release fertilizer possesses exceptional structural integrity and chemical stability during long-term storage.

3.8 | Water Holding and Retention Properties of SA-SRF in Soil

Structural characterization, water absorption, swelling properties, and pH responsiveness analysis of SA gels with varying ratios revealed that excessive acrylic acid (AA) content (e.g., St:AA = 2:1) enhances gel hydrophilicity (due to high carboxylate

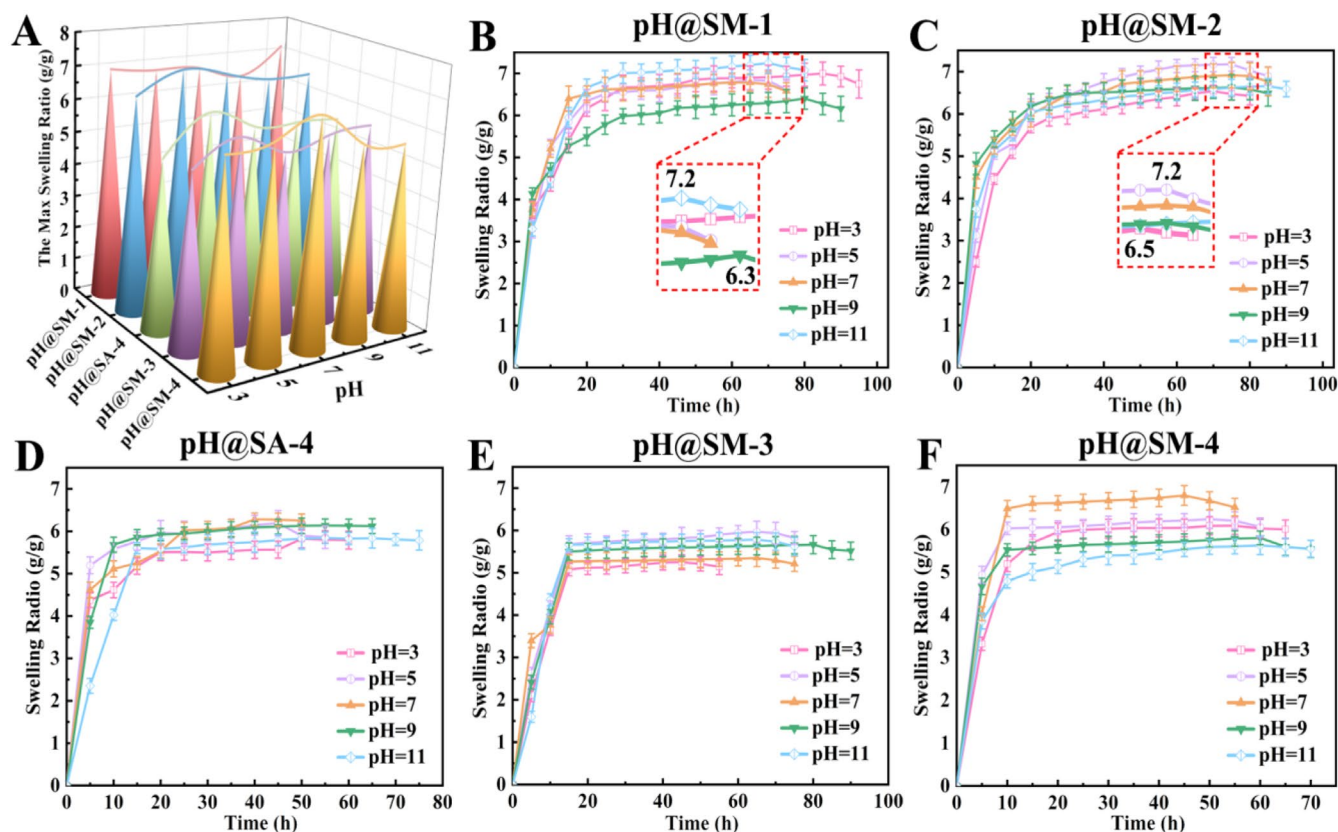


FIGURE 9 | Changes in equilibrium swelling rate response of SA gels with different MBA contents at different pH (A) and swelling rate change curves (B–F). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

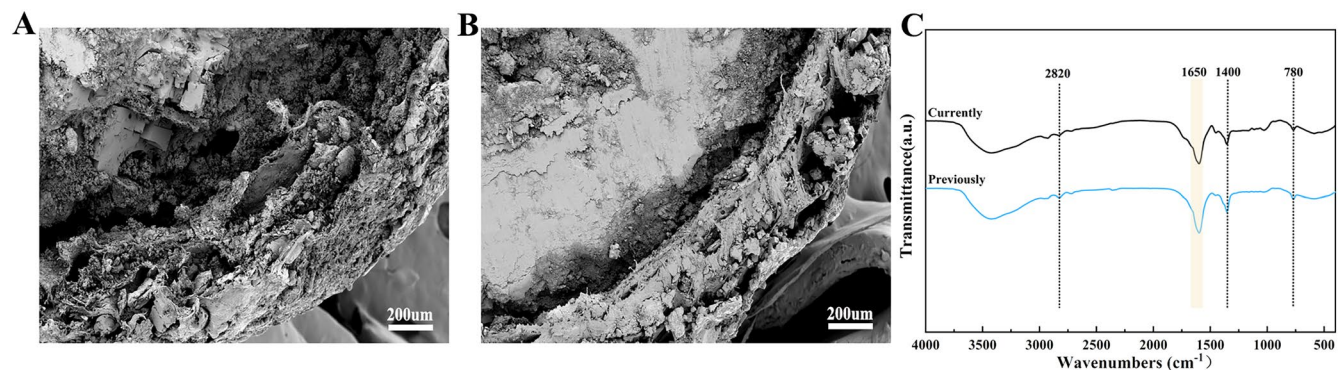


FIGURE 10 | SEM comparison of the sustained-release fertilizer gel layer (A) after storage and the newly prepared SA gel layer (B), and infrared spectrum comparison (C). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

group content) but reduces crosslinking density, thereby compromising structural integrity. SEM images demonstrated that pH@SA-1 exhibited excessively large pores and a loose structure. Conversely, the pH-responsive behavior of SA gels originates from protonation/deprotonation of AA's carboxylic acid ($-\text{COOH}$) groups. Insufficient AA content (e.g., St:AA = 5:1) diminishes pH responsiveness. Optimal performance was achieved at St:AA = 3:1, which exhibited the highest swelling ratio (7.85 g/g) and a uniform microstructure (SEM confirmed well-distributed pores in pH@SA-2), indicating a balance between responsiveness and mechanical stability. Regarding crosslinking agent selection, excessive MBA content (e.g., St:MBA = 300:1) caused over-crosslinking, restricting swelling

(pH@SM-1 swelling ratio: only 4.7 g/g). In contrast, insufficient MBA (e.g., St:MBA = 2000:1) resulted in weak crosslinking, leading to gel dissolution. The optimal St:MBA ratio of 500:1 provided structural stability (SEM revealed moderate pore size in pH@SM-2) while maintaining a high dissolution rate (6.9 g/g), indicating that this crosslinking density preserves gel integrity without compromising pH-responsive swelling. Therefore, to ensure optimal pH-responsive behavior and sustained-release properties of the fertilizer in weakly alkaline environments, SA gel prepared with a molar ratio of St to AA of 3:1 and St to MBA of 500:1 was selected as the coating material for the fertilizer to prepare pH-responsive slow-release fertilizer SA-SRF. After adding 1 g of SA-SRF to soils with different pH values, the water

retention capacity of SA-SRF was assessed, as illustrated in Figure 11A. In comparison to the soil lacking SA-SRF, the water retention of the reference soil was below 47%, while the water retention of SA-SRF increased by 3%–3.8% correspondingly. In a soil setting with pH = 7, following the addition of SA-SRF, the soil's water holding ability reached 30.29%, marking an 11.12% relative increase, as depicted in Figure 11B. The water retention rate of soil was 25.65%, 26.35%, 30.29%, 33.80%, and 31.28% after adding SA-SRF in each pH soil environment, compared with the control group in the neutral and alkaline soil at pH = 7 and 9, which effectively enhanced the water retention rate by about 10%, which shows the excellent water retention capacity

of SA-SRF in the soil environment, and the water evaporation could be effectively slowed down by adding SA-SRF.

3.9 | Slow and Controlled Release Performance of SA-SRF in Different pH Water and Soil Environments

In a water environment with pH = 7, pure urea was rapidly and wholly released within 24 h, and the cumulative release rate of nitrogen balance was 100.00%, as shown in Figure 12A, and the pure urea was released rapidly for the first 8 h in all

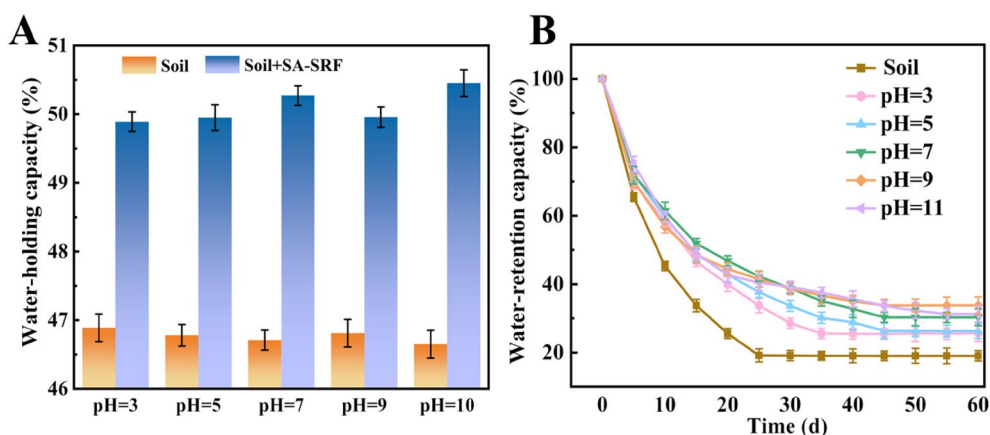


FIGURE 11 | Water holding and retention properties of SA-SRF in different pH soils (A, B). [Color figure can be viewed at wileyonlinelibrary.com]

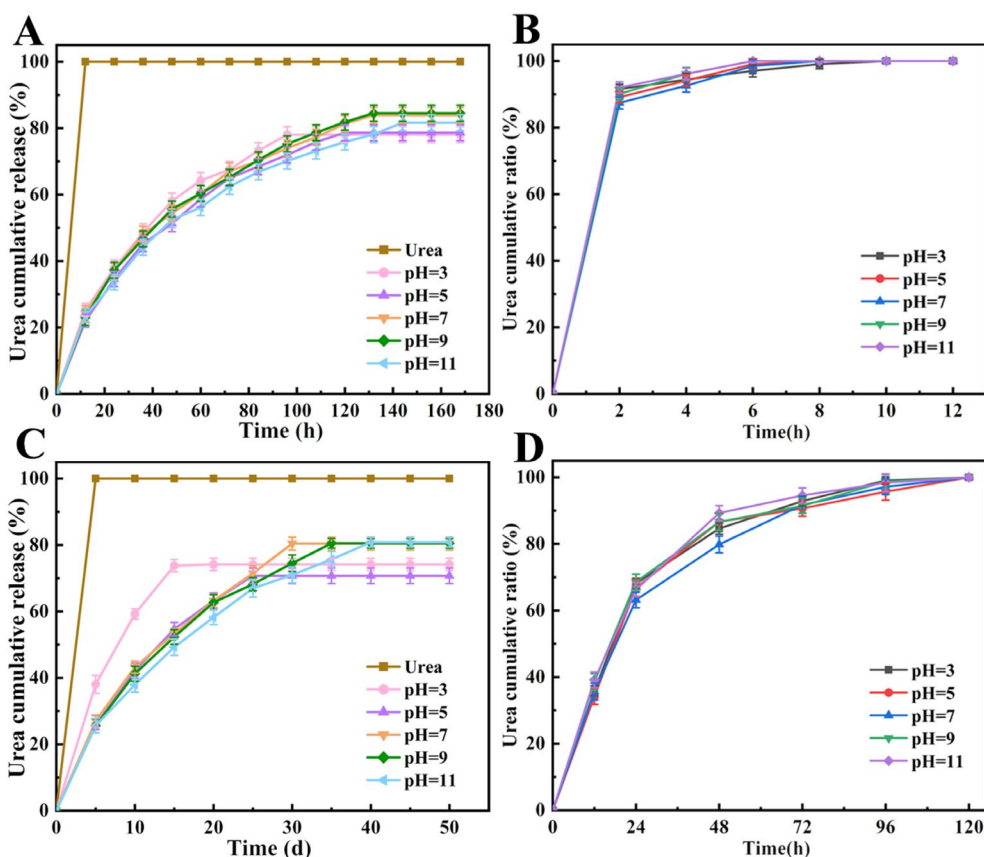


FIGURE 12 | Release behavior of SA-SRF (A) and normal urea fertilizer (B) in water of different pH and release behavior of SA-SRF (C) and normal urea fertilizer (D) in soil of different pH. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 | Fitted parameters for the kinetics of fertilizer release from SA-SRF slow-release fertilizers at different pHs.

pH value	Zero order model		First order model		Higuchi model		Korsmeyer-Peppas model		
	R^2	K_0	R^2	K_1	R^2	K_h	R^2	K_{kp}	n
3	0.4598	1.0308	0.9018	0.1558	0.7504	9.8401	0.9124	36.8079	0.1993
5	0.7408	1.3122	0.9354	0.0807	0.9339	11.2945	0.9558	18.8496	0.3764
7	0.7408	1.3672	0.9496	0.0807	0.9339	11.7675	0.9556	19.6391	0.3764
9	0.8009	1.4565	0.9604	0.0696	0.9628	12.2823	0.9719	17.0750	0.4191
11	0.8453	1.4533	0.9824	0.0616	0.9813	12.0701	0.9843	14.5341	0.4533

TABLE 3 | Various parameters of slow-release fertilizers in similar research work.

Materials	Response behavior	Fertilizer slow-release behavior	Water-retention in soil	References
SA-SRF	pH response	78.09% (pH = 7, 30 days) 80.83% (pH = 9, 35 days) 79.76% (pH = 11, 40 days)	30.29% (35 days) 33.80% (35 days) 31.28% (35 days)	This study
CMC/acrylamide/acrylic acid/HNTs	No	98.7% (6 days)	30.6% (6 days)	[33]
Cellulose/SA/MOFs/DMAA	pH response	50% (30 days)	0 (30 days)	[34]
CCEH-SRF	No	66.7% (35 days)	20.69 (7 days)	[35]
BSRNFs	No	75 (21 days)	No	[36]
SA/SX/OLS-SRF	No	77.08 (28 days)	22.31 (11 days)	[37]
OSHU	No	50.97 (35 days)	13.42 (16 days)	[38]
N,N-dimethyl-aminoethyl methacrylate/dopamine	pH and temperature response	54.13 (30 days)	No	[10]

pH ambient aqueous solutions with almost uniform release behavior, Figure 12B. The nitrogen equilibrium cumulative release rate of SA-SRF in various pH aqueous environments could reach about 80%, and the time required was 96, 120, 132, 132, and 144 h, respectively. Compared to urea pellets, SA-SRF in different pH aqueous environments with significant slow-release performance. Analysis of urea release behavior in soil environments (Figure 12C,D) revealed consistent rapid dissolution characteristics across all pH conditions, with no observable slow-release behavior. This rapid release profile results from direct urea-soil-water interactions, leading to nearly complete dissolution and equilibrium attainment within 96 h. Specifically, in neutral soil (pH=7), pure urea achieved complete release (100.00% nitrogen equilibrium) within 5 days. In contrast, the SA-SRF nitrogen release curve gradually reached equilibrium only after 30 days. Additionally, it is noteworthy that the fertilizer's release period can extend to 30–40 days in soil environments with pH levels of 7, 9, and 11. The nitrogen release rates in soils of these respective pH levels are 74.15%, 70.73%, 80.43%, 80.51%, and 80.85%. Compared to the equilibrium cumulative release rate of nitrogen in the water, the equilibrium cumulative release rate of nitrogen from SA-SRF in soil was less because the soil environment is more complex than the water environment, and the solubilization properties of SA and the slow-release

properties of SA-SRF are affected by more factors. By observing the slow-release curves in the two environments, we found that the PAA chain segments present in the gel coating layer of SA-SRF combined with the protons in the environment or the ionization effect affected the change of the pore space of the gel coating layer, which regulates the permeation process of the fertilizer medium. Thus, the infiltration process of the fertilizer medium was systematically regulated, enabling precise distribution and controlled release of nutrient components in water environments with varying pH levels and diverse soil conditions. This allows SA-SRF to demonstrate superior slow-release performance.

To analyze the responsive slow-release diffusion of fertilizers, kinetic modeling was performed using four release models. The Korsmeyer-Peppas model exhibited the highest correlation (R^2 , Table 2), suggesting a polymer diffusion-controlled nutrient release mechanism. All calculated n -values were below 0.5, consistent with Fickian diffusion. Neutral/alkaline conditions closely followed the Fickian model ($R^2 > 0.95$), while acidic conditions showed deviation from the model. Furthermore, swelling tests revealed pH-dependent variations in both equilibrium swelling rates and swelling kinetics across SA gel formulations. This consistency between swelling behavior and release profiles

confirms that nutrient diffusion is governed by differential swelling responses, suggesting pH-dependent alterations in the diffusion pathway of responsive fertilizers.

Comparative analysis of the data presented in Table 3 demonstrates that the SA-SRF developed in this study exhibits superior water retention capacity across soils with varying pH levels. Notably, the material shows particularly excellent controlled-release characteristics in alkaline to weakly alkaline soil conditions. Furthermore, the SA-SRF formulation satisfies all specifications for slow-release fertilizers as defined by both the European Committee for Standardization (CEN) [39] and the Chinese National Standard (GB/T 23348-2009) [40].

4 | Conclusion

In this work, we prepared a pH-responsive starch-based hydrogel-coated fertilizer, SA-SRF, using a disk coating method. The SA gel with molar ratios of St:AA of 3:1 and St:MBA of 500:1 showed optimal performance through structural and morphological characterization of the SA gel, water-absorbing and swelling properties, and pH-responsive properties. The SA-SRF prepared in this ratio effectively improved the soil water retention rate by about 10% or more in different pH soils, and showed excellent controlled release performance in various pH soil environments. In soil environments with pH = 7 and pH = 9, SA-SRF exhibits sustained-release periods of 30 and 35 days, respectively, with cumulative nitrogen release rates reaching 80.43% and 80.51%. Combining the results of structural characterization, water absorption and swelling performance, and pH response performance of SA, the fertilizer maintains good swelling performance in neutral and alkaline environments, and due to the presence of a large number of hydrophilic groups such as hydroxyl and carboxyl groups contained in the gel, it has the effect of absorbing and retaining a large amount of water. It can effectively improve the absorption of micronutrients in crops as well as their slow release, and it is suitable for the application in neutral and alkaline soil environments.

Author Contributions

Hui Li: conceptualization (lead), funding acquisition (lead), methodology (lead), project administration (lead), resources (lead), supervision (lead), writing – original draft (lead). **Shiyu Wang:** data curation (supporting), formal analysis (supporting), investigation (supporting), software (supporting), validation (supporting). **Xunhua Liao:** data curation (supporting), formal analysis (supporting), investigation (supporting), validation (supporting). **Haotan Yan:** investigation (supporting), software (supporting). **Jiaqi Luo:** investigation (supporting), resources (supporting), validation (supporting). **Peisong Li:** data curation (supporting), formal analysis (supporting), software (supporting). **Yangyang Li:** investigation (supporting). **Zhenbin Chen:** formal analysis (supporting), methodology (supporting), project administration (supporting), supervision (supporting), validation (supporting). **Hong Liu:** methodology (supporting), software (supporting), supervision (supporting), validation (supporting).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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